



Data Science with Python 3



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Chapter Four- DATA ACQUISITION AND PROCESSING-2

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OBJECTIVE

1. To familiarize with the Data Science Ethics
2. To introduce different aspects of privacy
3. To detect the future trends of data Science

4.1 DATA SCIENCE ETHICS

إن المهارة في استخراج الأنماط ذات الصلة في مجال علم البيانات تساعد في إحداث تغيير إيجابي في المجال الاجتماعي والتكنولوجي مما يجعلها **محايدة أخلاقياً**. فهي لا تأتي بمنظورها الخاص سواء: ما هو **صحيح** أو **غير صحيح**؛ أو: ما هو **جيد** أو **سيئ** في استخدامها. لا يوجد مثل هذا النوع من الإطار القائم على القيمة بينما الشركات التي تعمل على تخزين البيانات لديها نظام قائم على القيمة للتعامل مع المعلومات. أي شيء خاص أو محمي لا يمكن لأي شخص الوصول إليه باستثناء المسؤول أو عالم البيانات نفسه.

Analysts, data scientists, and information technology professionals must be concerned about **data science ethics**.

Anyone dealing with any type of data must **report any instances of data theft, unethical data collection, storage, use, etc.**

For example, from the first time a consumer enters their email address on your website to the time they purchase your goods, your organization may gather and keep data about their trips. People in the marketing team might be dealing with the data. The data of the person must be preserved.

مثلاً: من المرة الأولى التي يدخل فيها المستهلك عنوان بريده الإلكتروني على موقعك الإلكتروني حتى وقت شرائه للسلع، قد تجمع مؤسستك بيانات حول رحلاته وتحفظ بها. قد يتعامل الأشخاص في فريق التسويق مع البيانات. يجب الحفاظ على بيانات الشخص.

Protected data has been made public on the internet in the past, resulting in harm to persons whose information has been made available. **Misconfigured databases**, spyware, theft, or publishing on a public forum can all lead to **data leaks**.

لقد تم نشر البيانات المحمية على الإنترنت في الماضي، مما أدى إلى إلحاق الضرر بالأشخاص الذين تم نشر معلوماتهم. يمكن أن تؤدي قواعد البيانات غير المهيئة أو برامج التجسس أو السرقة أو النشر على منتدى عام إلى تسريب البيانات.

Individuals and organizations must use safe computing practices, conduct frequent system audits, and adopt policies to address computer and data security.

Companies must take appropriate cybersecurity steps to prevent the leakage of data and information. This is more important for banks and financial institutions which deal with customers' money. Protections must be maintained even when equipment is transferred or disposed of, according to policies.

يتعين على الأفراد والمؤسسات استخدام ممارسات الحوسبة الآمنة، وإجراء عمليات تدقيق متكررة للنظام، وتبني سياسات لمعالجة أمن الكمبيوتر والبيانات. ويتعين على الشركات اتخاذ خطوات الأمن السيبراني المناسبة لمنع تسرب البيانات والمعلومات. وهذا أكثر أهمية بالنسبة للبنوك والمؤسسات المالية التي تتعامل مع أموال العملاء. ويجب الحفاظ على الحماية حتى عند نقل المعدات أو التخلص منها، وفقًا للسياسات.

4.2 DATA SCIENCE – GOOD ASPECTS FOR TECHNOLOGY

To acclimatize the organization with the ethical data science, there's a need to understand what actually are the ethics about data science with the cost requirement for the implementation of ethical values and what can be the ways to execute such practices.

لتأقلم المنظمة مع علم البيانات الأخلاقي، هناك حاجة لفهم ما هي الأخلاقيات الفعلية حول علم البيانات مع متطلبات التكلفة لتنفيذ القيم الأخلاقية وما هي الطرق التي يمكن أن تكون لتنفيذ مثل هذه الممارسات.

There are several ways and means with the respective solutions for the same.

Firstly, the data scientist needs to understand about the daintiness of the information in the data store. With this the relevant implementation or the operations can be performed on the data keeping in the view the consequences with improper or unacceptable access of the data for information. Indeed, the security perspective in

computer or network has proved many times that ignoring the consequences of accessing the data unethically might cause a trouble to the data professionals or the company as a whole in terms of loss of reputation or money.

أولاً، يحتاج عالم البيانات إلى فهم دقة المعلومات الموجودة في مخزن البيانات. وبهذا يمكن التنفيذ أو العمليات ذات الصلة على البيانات مع مراعاة العواقب المترتبة على الوصول غير السليم أو غير المقبول إلى البيانات للحصول على المعلومات. والواقع أن منظور الأمن في الكمبيوتر أو الشبكة أثبت مرات عديدة أن تجاهل عواقب الوصول إلى البيانات بطريقة غير أخلاقية قد يسبب مشكلة لمحترفي البيانات أو الشركة ككل من حيث خسارة السمعة أو المال.

With the available systems it is difficult for any data scientist to predict the inevitable unethical uses of the data but with the AI based techniques especially using machine learning and deep learning, a prediction can be made for the unintended consequences. For example, Facebook's "Year in Review" that reminded people of deaths and other painful events. This can be handled by the data professionals in the field of data science by keeping in view the patterns of the data to be explored and how to think of better means to represent the data with a new or enhanced approach.

مع الأنظمة المتاحة، من الصعب على أي عالم بيانات التنبؤ بالاستخدامات غير الأخلاقية الحتمية للبيانات، ولكن باستخدام تقنيات الذكاء الاصطناعي، وخاصة باستخدام التعلم الآلي والتعلم العميق، يمكن التنبؤ بالعواقب غير المقصودة. على سبيل المثال، "مراجعة العام" على فيسبوك التي ذكرت الناس بالوفيات والأحداث المؤلمة الأخرى. يمكن لمحترفي البيانات في مجال علوم البيانات التعامل مع هذا من خلال وضع أنماط البيانات المراد استكشافها في الاعتبار وكيفية التفكير في وسائل أفضل لتمثيل البيانات بنهج جديد أو محسن.

Another important step to stop the data mining if the administrator finds any kind of problem in the production line. This idea goes back to Toyota's Kanban: any assembly line worker can stop the line if they see something going wrong. The line doesn't restart until the problem is fixed. Workers don't have to fear consequences from management for stopping the line; they are trusted and expected to behave responsibly.

وهناك خطوة أخرى مهمة لوقف التنقيب عن البيانات إذا وجد المسؤول أي نوع من المشاكل في خط الإنتاج. وتعود هذه الفكرة إلى كانبان تويوتا: أي عامل في خط التجميع يمكنه إيقاف الخط إذا رأى شيئاً ما يحدث خطأ. ولا يستأنف الخط العمل حتى يتم إصلاح المشكلة. ولا يتعين على العمال أن يخشوا العواقب من الإدارة لإيقاف الخط؛ فهم موضع ثقة ومن المتوقع أن يتصرفوا بمسؤولية.

The issue lurking behind all of these concerns is, of course, corporate culture. Corporate environments can be hostile to anything other than short-term profitability. That's a consequence of poor court decisions and economic doctrine, particularly in the U.S. But

that inevitably leads us to the biggest issue: how to move the needle on corporate culture. Susan Etlinger has suggested that, in a time when public distrust and disenchantment is running high, ethics is a good investment. Upper-level management is only starting to see this; changes to corporate culture won't happen quickly. Users want to engage with companies and organizations they can trust not to take unfair advantage of them. Users want to deal with companies that will treat them and their data responsibly, not just as potential profit or engagement to be maximized. Those companies will be the ones that create space for ethics within their organizations. We, the data scientists, data engineers, AI and ML developers, and other data professionals, have to demand change. We can't leave it to people that "do" ethics.

We can't expect management to hire trained ethicists and assign them to our teams. We need to live ethical values, not just talk about them. We need to think carefully about the consequences of our work. We must create space for ethics within our organizations. Cultural change may take time, but it will happen—if we are that change. That's what it means to do good data science.

4.3 OWNERS OF DATA

Data owners are either individuals or teams who make decisions such as who has the right to access and edit data and how it's used. Owners may not work with their data every day, but are responsible for overseeing and protecting a data domain.

A data owner is responsible for the data in a particular data domain. They may belong to the steering committee and ensure that the data under their view is governed

throughout the organization. Data owners approve data glossaries and definitions as well as initiate data quality activities.

Owning' data – who owns data, what's capable of being owned, and what rights and responsibilities ownership attracts – is gaining a lot of attention.

4.3.1 Responsibilities of the Data Owner

The first responsibility of the data owner is to classify the data correctly. Once a classification has been set it is up to the data owner to determine who has access to the data.

Usually, this access is based upon roles as opposed to individuals.

The data classification is one of the most important steps. Data classification has a different meaning for different organizations but at the basic level it is knowing the type of data a company has, determining its value, and categorizing it. For example, if your company has a secret sauce or original intellectual property it may be considered “top secret” or “confidential”. The reason to label or classify it as “top secret” or “confidential” is so it can be handled and ultimately protected appropriately. In addition to not putting the correct controls on data there is the potential to retain data for longer than needed or destroy data before it should be based on laws or contractual commitments. Many people have an opinion on how data should be classified or labeled but at the end of the day it is the responsibility of the data owner who is ultimately accountable for the data to make the final decision. The data owner will have the most knowledge of the use of the data and the value to the company. It is advisable for the data owner to get input from various sources like the data custodian or data users but the data owner has complete control over the data.

- Who has access to the data? Clarify the roles of people who can access the data.

Example: Employees can see an organization chart with departments, manager names, and titles but not salary information (Classification = internal). But a very limited audience like HR should only have access to salary data, performance data, or social security numbers (Classification = confidential).

- How is the data secured? Sensitive data elements within HR documentation have been classified to be confidential and therefore it requires additional security controls to protect it. Some of the additional controls to secure confidential data stored in electronic medium could include being saved in a location on the network with appropriate safeguards to prevent unauthorized access (secure folders protected by passwords).

- How long the data is retained? Many industries require that data be retained for a certain length of time. For example, the finance industry requires a seven-year retention period and some health care industries requires a 100-year retention period. Data owners need to know the regulatory requirements for their data and if there is no clear guidance on retention then it should be based off the company's retention policy or best practices.

- How data should be destroyed? Based on the classification of the data there should be clear guidance on how to dispose or destroy the data. For example:

Information Medium	Public	Internal	Confidential
Hard Copy	Place in recycling bins or trash receptacles.	Place in secured shedding bins or manually shred.	Place in secured shedding bins or manually shred with a cross-cut shredder or pulped. A record must be maintained that indicates the records disposed of and the data of disposal.
Electronic records	Electronic records can be deleted normally.	Electronic records need to be overwritten to 1's and 0's or with a secure delete option.	Electronic records need to be degaussed off magnetic media after securely deleted or the physical media should be destroyed with a record maintained that indicates the records disposed of and the date of disposal.

- What data needs to be encrypted? Data owners should decide whether their data needs to be encrypted. To make this determination the data owner should know the applicable laws or regulation requirements set that must be complied with. A good example of a regulation requirement is set by the Payment Card Industry (PCI) Data Security Standard and it requires that the transmission of cardholder data across open, public networks must be encrypted.

4.3.2 The Importance of Assigning Data Owners

In most organizations, as data passes through different teams and systems, assigning data owners can be cumbersome. However, this is a critical step for GDPR compliance.

Why assigning data owners is important:

1. Accountability — Ownership creates accountability. Since GDPR introduces many controls on personal data, assigning responsibilities ensures that data will be continuously -monitored for compliance by the owners.

2. Defining policies — As they have a vested interest in the integrity of their data, owners focus on defining policies (for example retention or deletion policies) and standards that ensure the alignment of their data to the GDPR.

3. Creating trusted data — Data ownership is a key ingredient to gain customer trust and achieve measurable business benefits. Poor data could easily result in bad customer experiences and ultimately losing customers. In particular, when personal data is not reconciled into a data subject 360° view, compliance with data subject access rights, such as rights of portability or rights to be forgotten cannot be fully achieved.

4. Eliminating redundancies — As organizations strive to put the appropriate governance framework in place for GDPR, one common frustration is a loss of productivity. This issue stems from multiple teams addressing the same problem either because there isn't a clear understanding of data or they're not even aware that the problem has been resolved by another team. Federated ownership eliminates these painful issues.

4.3.3 Identification of Data Owners: Three Questions to Ask

The mandatory introduction of a data protection officer (DPO) role by GDPR in most organizations effectively creates a master data owner. Each and every element in a data taxonomy needs an individual owner, however, and there is little likelihood that a single

DPO can hold this responsibility on such a large scale. In addition, this could create a security issue. Delegation and segregation of duties are needed.

Asking the right questions helps. Once these questions have been answered, the data owner should become clearer:

1. Who is most impacted by data accuracy?
2. Who has authority to decide the next step?
3. Who owns the related data attributes?

4.4 DIFFERENT ASPECTS OF PRIVACY

First there is the data where security and privacy has always mattered and for which there is already an existing and well galvanized body of law in place. Foremost among these is classified or national security data where data usage is highly regulated and enforced.

Other data for which there exists a considerable body of international and national law regulating usage includes:

Proprietary Data – specifically the data that makes up the intellectual capital of individual businesses and gives them their competitive economic advantage over others, including data protected under copyright, patent, or trade secret laws and the sensitive, protected data that companies collect on behalf of its customers;

Infrastructure Data - data from the physical facilities and systems – such as roads, electrical systems, communications services, etc. – that enable local, regional, national, and international economic activity; and

Controlled Technical Data - technical, biological, chemical, and military-related data and research that could be considered of national interest and be under foreign export restrictions.

It may be possible to work with publicly released annualized and cleansed data within these areas without a problem, but the majority of granular data from which significant insight can be gleaned is protected. In most instances, scientists, researchers, and other authorized developers take years to appropriately acquire the expertise, build the personal relationships, and construct the technical, procedural and legal infrastructure to work with the granular data before implementing any approach. Even using publicly released datasets within these areas can be restricted, requiring either registration, the recognition of or affiliation to an appropriate data governing body, background checks, or all three before authorization is granted. The second group of data that raises privacy and security concerns is personal data. Commonly referred to as Personally Identifiable Information (PII), it is any data that distinguishes individuals from each other. It is also the data that an increasing number of digital approaches rely on, and the data whose use tends to raise the most public ire. Personal data could include but is not limited to an individual's:

- Government issued record data (social security numbers, national or state identity numbers, passport records, vehicle data, voting records, etc.);
- Law enforcement data (criminal records, legal proceedings, etc.);
- Personal financial, employment, medical, and education data;
- Communication records (phone numbers, texts data, message records, content of conversations, time and location, etc.);

- Travel data (when and where traveling, carriers used, etc.);
- Networks and memberships (family, friends, interests, group affiliations, etc.);
- Location data (where a person is and when);
- Basic contact information (name, address, e-mail, telephone, fax, twitter handles, etc.);
- Internet data (search histories, website visits, click rates, likes, site forwards, comments, etc.);
- Media data (which shows you're watching, music you're listening to, books or magazines you're reading, etc.);
- Transaction data (what you're buying or selling, who you're doing business with, where, etc.); and
- Bio and activity data (from personal mobile and wearable devices).

In industries where being responsible for handling highly detailed personal data is the established business norm – such as in the education, medical and financial fields – there are already government regulations, business practices and data privacy and security laws that protect data from unauthorized usage, including across new digital platforms. But in many other industries, particularly in data driven industries where personal data has been treated as proprietary data and become the foundation of business models, there is currently little to no regulation. In the new normal, the more that a data approach depends on data actively or passively collected on individuals, the more likely that consumers will speak up and demand

privacy protection, even if they previously gave some form of tacit approval to use their data.

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Despite this new landscape, there are lots of different ways to use personal data, some of which may not trigger significant privacy or security concerns. This is particularly true in cases where individuals willingly provide their data or data cannot be attributed to an individual. Whether individuals remain neutral to data approaches tends to be related to the level of control they feel they have over how their personal data is used. Some organizations that collect personal data extensively, such as Facebook and Google, work to increasingly provide their users with methods to control their own data. But for others, the lack of due diligence on data privacy in their approaches has already had their effect.

A third category of data needing privacy consideration is the data related to good people working in difficult or dangerous places. Activists, journalists, politicians, whistle-blowers, business owners, and others working in contentious areas and conflict zones need secure means to communicate and share data without fear of retribution and personal harm.

4.5 FIVE C' S OF DATA SCIENCE

What does it take to build a good data product or service? Not just a product or service that's useful, or one that's commercially viable, but one that uses data ethically and responsibly.

Users lose trust because they feel abused by malicious ads; they feel abused by fake and misleading content, and they feel abused by —act first, and apologize profusely later|| cultures at many of the major online companies. And users ought to feel abused by many abuses they don't even know about. Why was their insurance claim denied? Why weren't they approved for that loan? Were those decisions made by a system that was trained on biased data? The slogan goes, "Move fast and break things." But what if what gets broken is society?

Data collection is a big business. Data is valuable: "the new oil," as the Economist proclaimed. We've known that for some time. But the public provides the data under the assumption that we, the public, benefit from it. We also assume that data is collected and stored responsibly, and those who supply the data won't be harmed. Essentially, it's a model of trust. But how do you restore trust once it's been broken? It's no use pretending that you're trustworthy when your actions have proven that you aren't. The only way to get trust back is to be trustworthy, and regaining that trust once you've lost it takes time.

There's no simple way to regain users' trust, but a —golden rule|| for data as a starting point: —treat others' data as data scientist would like other to treat their data.|| However, implementing a "golden rule" in the actual research and development process is challenging.

The golden rule isn't enough by itself. There has to be certain guidelines to force discussions with the application development teams, application users, and those who

might be harmed by the collection and use of data. Five framing guidelines help us think about building data products. We call them the five **Cs**: **consent**, **clarity**, **consistency**, **control** (and transparency), and **consequences** (and harm).

4.5.1 Consent

The trust between the people who are providing data and the people who are using it cannot be established without agreement about what data is being collected and how that data will be used . Agreement starts with obtaining consent to collect and use data. Unfortunately, the agreements between a service's users (people whose data is collected) and the service itself (which uses the data in many ways) are binary (meaning that you either accept or decline) and lack clarity. In business, when contracts are being negotiated between two parties, there are multiple iterations (redlines) before the contract is settled. But when a user is agreeing to a contract with a data service, you either accept the terms or you don't get access. It's non-negotiable.

Data is frequently collected, used, and sold without consent. This includes organizations like Acxiom, Equifax, Experian, and Transunion, who collect data to assess financial risk, but many common brands also collect data without consent. In Europe, Google collected data from cameras mounted on cars to develop new mapping products. AT&T and Comcast both used cable set top boxes to collect data about their users, and Samsung collected voice recordings from TVs that respond to voice commands.

4.5.2 Clarity

Clarity is closely related to consent. Users must have clarity about what data they are providing, what is going to be done with the data, and any downstream consequences of how their data is used. All too often, explanations of what data is collected or being sold are buried in lengthy legal documents that are rarely read carefully, if at all. Observant readers of Eventbrite's user agreement recently discovered that listing an event gave the company the right to send a video team, and exclusive copyright to the recordings. And the only way to opt out was by writing to the company. The backlash was swift once people realized the potential impact, and Eventbrite removed the language.

Facebook users who played Cambridge Analytica's —This Is Your Digital Life|| game may have understood that they were giving up their data; after all, they were answering questions, and those answers certainly went somewhere. But did they understand how that data might be used? Or that they were giving access to their friends' data behind the scenes? That's buried deep in Facebook's privacy settings. It really doesn't matter which service you use; you rarely get a simple explanation of what the service is doing with your data, and what consequences their actions might have. Unfortunately, the process of consent is often used to obfuscate the details and implications of what users may be agreeing to. And once data has escaped, there is no recourse.

4.5.3 Consistency and Trust

Trust requires consistency over time. You can't trust someone who is unpredictable.

They may have the best intentions, but they may not honor those intentions when you need them to. Or they may interpret their intentions in a strange and unpredictable way.

And once broken, rebuilding trust may take a long time. Restoring trust requires a prolonged period of consistent behavior.

Consistency, and therefore trust, can be broken either explicitly or implicitly. An organization that exposes user data can do so intentionally or unintentionally. In the past years, we've seen many security incidents in which customer data was stolen: Yahoo!, Target, Anthem, local hospitals, government data, and data brokers like Experian, the list grows longer each day. Failing to safeguard customer data breaks trust—and safeguarding data means nothing if not consistency over time.

4.5.4 Control and Transparency

All too often, users have no effective control over how their data is used. They are given all-or-nothing choices, or a convoluted set of options that make controlling access overwhelming and confusing. It's often impossible to reduce the amount of data collected, or to have data deleted later. A major part of the shift in data privacy rights is moving to give users greater control of their data. For example, Europe's General Data Protection Regulation (GDPR) requires a user's data to be provided to them at their request and removed from the system if they so desire.

4.5.5 Consequences

Data products are designed to add value for a particular user or system. As these products increase in sophistication, and have broader societal implications, it is essential to ask whether the data that is being collected could cause harm to an individual or a group. The unforeseen consequences and the —unknown unknowns|| about using data and combining data sets have been witnessed frequently. Risks can

never be eliminated completely. However, many unforeseen consequences and unknown unknowns could be foreseen and known, if only people had tried. All too often, unknown unknowns are unknown because we don't want to know.

While Strava and AOL triggered a chain of unforeseen consequences by releasing their data, it's important to understand that their data had the potential to be dangerous even if it wasn't released publicly. Collecting data that may seem innocuous and combining it with other data sets has real-world implications. It's easy to argue that Strava shouldn't have produced this product, or that AOL shouldn't have released their search data, but that ignores the data's potential for good. In both cases, well-intentioned data scientists were looking to help others. The problem is that they didn't think through the consequences and the potential risks.

Many data sets that could provide tremendous benefits remain locked up on servers.

Medical data that is fragmented across multiple institutions limits the pace of research. And the data held on traffic from ride-sharing and gps/mapping companies could transform approaches for traffic safety and congestion. But opening up that data to researchers requires careful planning

4.6 DIVERSITY – INCLUSION

Reports of AI gone wrong abound, and Responsible AI has started to take a foothold in business — Gartner has even added Responsible AI as a new category on its Hype Cycle for Emerging Technologies. Yet when talking about solutions, increasing diversity and making data science a more inclusive field unfortunately don't often top the list. Noelle

Silver, Head of Instruction, Data Science, Analytics, and Full Stack Web Development at HackerU, is looking to change that.

A 2018 study revealed that only 15% of data scientists are women, and sadly, a 2020 study found exactly the same results: it seems we haven't managed to move the needle. While diversity obviously encompasses more than just women, few studies have been able to quantify other types of representation in the field. Inclusivity is similarly difficult to quantify, both in terms of people working on technology and the ways in which technology can be accessed by all. But that doesn't mean there aren't solutions.

Problem

“The reality is that when we train machine learning models with a bunch of data, it's going to make predictions based on that data. If that data comes from a room of people that look the same, talk the same, act the same, they're all friends — it's not a bad scenario. In the moment, you feel like things are good. No one is really seeing any problems; you don't feel any friction. It's very misleading, especially in artificial intelligence. So, you go to market.

The problem, though, is not everyone looks like you, or talks like you, or thinks like you. So even though you found a community of people that built this software that thinks the same, as soon as you go to market and someone other than that starts using it, they start to feel that friction.”

Solution

Of course, there's no easy, magic bullet solution to this problem, but foundations of a good start are:

- Committing the time and resources to practice inclusive engineering: This includes, but certainly isn't limited to, doing whatever it takes to collect and use diverse datasets.
- Create an experience that welcomes more people to the field: This might mean looking at everything from education to hiring practices.
- Think beyond regulations: Simply being compliant doesn't necessarily mean experiences are optimized.